

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

FLEETMATE: Car Rental System

In partial fulfilment for the requirements in
Capstone 1 and Research Subject

Bonnie Boy F. Franco III

Ejvind Gem Gimotea

John Mic Khael Sumbing

Josh Bretaña

John Aaron Alegarbes

System Development

Second Semester

2025-2026

CHAPTER 1

INTRODUCTION

1.1 Background of the study

Many small vehicle rental businesses still manage bookings, customer records, and vehicle availability manually. To manage these tasks, owners often use messaging platforms to communicate with customers and a note-taking app to record bookings and customer details. While this might work at first due to having a small number of transactions, they will become increasingly difficult to keep track as the number of customers grows.

Based on the interview with the owner of JR4L Rental Service, Rex Quiñon, the current rental process is handled using a notes app to record customer details and rental dates. Messaging platforms are used to communicate with customers, and a few AI tools to sometimes answer customer's inquiries, but with the owner's limited knowledge of digital platforms makes it hard to put many functionalities into practice.

Customers usually ask about the availability, pricing, and the rental schedules of the vehicles, and they expect quick responses and confirmations. Without a system to manage such inquiries, they must message the owner directly to check if a vehicle is available on specific dates, or how much is the rental. This also requires the owner to check their records for the availability of the vehicle before replying to the customer which can slow down the booking process or can accidentally cause double booking or schedule conflicts.

This capstone project proposes the development of **FleetMate**, designed to be a digital rental management system to help the business organize booking schedules, manage customer information, and monitor availability more efficiently. The system aims to reduce manual workload, prevent schedule conflicts, and improve response time, ultimately improving the efficiency and overall operations of the business

1.2 Statement of the Problem

Despite the availability of many applications for tracking information, many small vehicle rental businesses still rely on manual methods to handle their business. This study seeks to address the following problems:

1. Vehicle availability is not tracked clearly, which can lead to double bookings and scheduling conflicts.
2. Rental requests, active rentals, and return schedules are hard to monitor due to the lack of a centralized system.
3. Customer information and rental records are managed manually using a note app, making data difficult to organize and retrieve.
4. Manual record checking causes delays in responding to customer inquiries about availability and rental detail.
5. Lack of centralized customer records making it difficult to access client information.
6. Inefficient managing of operational processing due to lack of an integrated system.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop a car rental system that helps organize booking schedules, manage customer information and track availability to improve efficiency of vehicle rental management.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. **To analyze** the current rental management process used by JR4L Rental Services and identify the limitations of the manual system
2. **To design** a user-friendly interface for managing vehicle bookings, schedules, and customer information.
3. **To develop** the FleetMate rental management system to organize booking records and rental transactions.
4. **To implement** features that monitor vehicle availability, active rentals, and return schedules.
5. **To integrate** a centralized database for storing and managing customer and rental information.
6. **To evaluate** the system's functionality, usability, and performance through system testing.

1.4 Scope and Delimitation

This study focuses on the development of a digital rental management system that will help small vehicle rental businesses in managing their rental operations. The system is designed to help organize booking schedules, and manage customer information in an efficient and simple manner to prevent double bookings and scheduling conflicts.

The system is limited to the development and implementation of the system for JR4L Rental Service. It will focus on improving the recording and monitoring of business operations. Other problems from other small rental services businesses will not be accounted for and only aimed towards JR4L Rental Services. Other features like online payment will not be included in the system.

1.5 Significance of the Study

The proposed system will benefit the following:

- **Business Owner** – the system will provide a centralized system to manage vehicle availability, rental schedules, and customer records more efficiently, reducing manual workload and scheduling conflicts.
- **Customers** – the system will enable faster responses to inquiries and clearer information about vehicle availability and rental schedules
- **BSIT Students** – by serving as a learning tool for web development, database management, and system design
- **Future Researchers**– this study may serve as a reference for future studies or improvements related to rental management systems, filtering, and request approval management.

1.6 Definition of Terms

Booking – The process of reserving a vehicle for a specific date and time.

Centralized System – A system where all data and processes are stored and managed in a single location, allowing easier access, organization, and control of information.

Database – An organized collection of data used by the system to store information such as customer details, vehicle records, and rental transactions.

Fleet Management - the process of overseeing and coordinating a group of vehicles used by a business to ensure they operate efficiently, safely, and cost-effectively.

Rental Management System – A digital system designed to manage rental operations such as bookings, vehicle availability, and customer records.

Rental Record – Information stored in the system that includes customer details, vehicle used, rental dates, and return schedules.

Web-Based System – A software application that can be accessed through a web browser using the internet.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Foreign studies show the increasing role of web-based technologies in improving efficiency and reliability of car rental operations. Based on **Gupta et al. (2024)** modern online vehicle rental platforms that uses frontend and backend technologies such as React.Js for the frontend, and Node.JS and Express.js for backend can help user experience through secure authentication and real-time system performance. Their findings revealed that system that implement such functions are able to handle high user demand while maintaining usability as reflected in high user ratings.

Similarly, **Divya Vanil et al. (2025)** developed an integrated car booking and rental platform designed to make car rental operations more organized and efficient through a web-based system. The study focused features such as vehicle management, customer booking, maintenance monitoring, and role-based access for administrators, employees, and customers. Their findings show that combining the functions into single platform helps improve record management, simplify rental processes, and provide reliable service for both businesses and customers.

#####**Also Dhoke et al. (2024)** emphasized the importance of implementing fleet monitoring and real-time booking systems in car rental platforms. Their researched showed that centralized dashboards can reduce manual processes and improve overall service quality. By providing real-time data access, businesses are able to make faster and more informed decisions regarding vehicle availability and customer service.#####

#####Lastly, **Jyothi and Sheethal (2025)** highlighted those key features such as, real-time vehicle status updates, automated notifications, can further enhance operational efficiency. Their study found that systems supporting both web and mobile platforms promote seamless interaction between users and service provider thus improving customer satisfaction and operations of the business#####

2.2 Local Studies

Local studies show how car rental systems in the Philippines are being improved by integrating modern system features and digital solutions. According to **Albino and Acebedo**, the development of a car rental management system tailored for rental businesses in Zambales incorporated a scheduling algorithm to manage car reservation and vehicle availability. This shows that user-friendly and responsive systems can improve efficiency and make rental operations easier to manage.

Similarly, De Jesus et al. (2021) addressed common issues in car rental businesses such as booking inefficiencies, difficulty in monitoring reservations, and challenges in vehicle tracking through the development of a web-based booking and management system with a mobile tracking application. The study highlighted features such as reservation management, customer monitoring, and vehicle tracking, which helped improve operational efficiency and service reliability. Their findings emphasized that integrating these functions into a centralized digital platform can simplify rental operations and improve the overall management of rental services.

Moreover, broader analyses of car rental operations in the Philippine highlighted persistent challenges such as inconsistent vehicle availability issues, reservation errors, maintenance tracking, and fuel level management. These findings emphasize the need for integrated digital solutions that enhance accessibility, streamline operations, and improve overall customer satisfaction. By using technology, rental businesses can address these issues and achieve more efficient and reliable service management.

2.3 Related Literature

Information Systems Theory and Development Concepts

Systems Development Life Cycle (SDLC) The SDLC provides a structured framework for information systems development, typically comprising five phases: planning and problem identification, analysis, design, development, and operation and maintenance. Avison et al. (2006) highlight SDLC's simplicity, methodological basis, and formal task division between business and technical specialists. The traditional waterfall model, while sequential and rigid, ensures comprehensive documentation and phase-specific deliverables.

Agile Methodologies Agile models emphasize incremental development, cooperative client-team communication, simplicity, and adaptability to changing requirements. These methodologies have become widely accepted for organizational product development, offering flexibility compared to traditional rigid approaches.

Software Development Technologies

Web-Based Application Architecture Modern web-based systems typically employ three-tier architecture: frontend presentation layer (HTML, CSS, JavaScript frameworks like React.js), backend application layer (Node.js, Express.js, or PHP), and database layer (MySQL, MongoDB). This separation of concerns enhances maintainability and scalability.

Database Management Systems Relational databases (MySQL) and NoSQL databases (MongoDB) serve as the foundation for rental management systems. MySQL provides structured data management with ACID compliance, while MongoDB offers flexibility for unstructured data and horizontal scalability.

User Interface Design Principles Research emphasizes the importance of user-friendly interfaces for non-technical users, including intuitive navigation, responsive design, and clear visual hierarchies. Systems achieving high usability ratings (92% user satisfaction) prioritize simplicity and streamlined booking processes.

Fleet Management and Optimization

Fleet Management Software (FMS) Fleet management systems function as centralized dashboards tracking vehicle data including speed, temperature, fuel level, geographical location, and route direction. Modern FMS integrates GPS technology, sensors, and IoT devices to provide real-time visibility and operational insights.

Operational Efficiency Metrics Studies demonstrate that digital fleet solutions can reduce logistic overhead by 27-36%, increase vehicle profitability, and optimize route planning. Key features include automated maintenance notifications, real-time tracking, and comprehensive reporting tools.

2.4 Synthesis of the Reviewed Literature

Summary of Findings

The reviewed foreign and local studies show the importance of web-based rental systems both globally and locally. These systems focus on improving operational efficiency and service management in the car rental business. Foreign studies highlight the benefits of digital platforms in reducing manual work, minimizing errors, and enhancing customer satisfaction. These platforms allow businesses to streamline key operations such as online reservations, fleet monitoring, and real-time vehicle availability within a single system, resulting in more efficient service delivery.

Local studies further support these findings by showing that many rental businesses still rely on manual processes, which lead to inefficiencies, scheduling conflicts, and inaccurate record-keeping. Foreign studies focus on advanced technologies and integrated features, while local studies mainly address common operational problems such as manual record-keeping and scheduling conflicts. Although existing studies show the effectiveness of web-based rental systems, many small rental businesses still lack a centralized and affordable platform for managing bookings, vehicle availability, and customer records.

These studies serve as the foundation for the development of FleetMate, which aims to provide a centralized and user-friendly rental management system for small vehicle rental businesses. The proposed system seeks to address these problems by providing efficient functions for booking management, vehicle tracking, and organized information handling.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study will use a developmental research approach to develop a web-based car rental system platform for JR4L Rental Service. This design focuses on the creation, design, and evaluation of the system intended to improve the efficiency of business operations.

3.2 System Development Methodology

This study will use the **Agile Development Model**, consisting of the following phases:

1. Requirements Analysis

- Information will be gathered through interviews with the owner to understand the current system and what area to improve

2. System Design

- The structure of the system, database design, and User Interface will be planned based on the requirements

3. Development

- The system will be built on React, Supa Base, PostgreSQL

4. Testing

- The testing will be to identify errors, bugs, any unintentional functions that might occur, to make sure that the proper functionality is working, and to verify that it meets the requirements

5. Deployment (Prototype)

- The system will be deployed for use by the client for managing rental operations and for users to use to book rentals

6. Continuous Improvement/Maintenance

- Future improvements and updates will be applied to enhance system performance

3.3 System Architecture

FleetMate will follow a client-server architecture. Where the users interact with the system through a web or mobile interface, while the server handles all processes and database management.

Users will handle user interaction, like input of rental information, viewing availability, and navigation of system features through a web or mobile interface, while the server will process user requests, perform necessary operations, and will communicate with the database to store and retrieve data such as customer records, booking schedules, and vehicle status.

The administrator will log in to the system to manage vehicle listings, customer records, and rental schedules. Data entered by the user will be processed by the system and will be stored in a centralized database. The system will then retrieve and display updated information such as vehicle availability and booking status.

3.4 Tools and Technologies Used

Frontend: React

Backend: Supa Base

Database: PostgreSQL

Development Tools: Visual Studio Code, Git/GitHub,

3.5 Data Gathering Techniques

- Interviews-

Interviews were conducted with the owner of JR4L Rental Service (Rex Quiñon) to gather information about how the current process was doing. This process helped identify the challenges in managing bookings, keeping customer records, and checking vehicle availability. -##### Information gathered were used as a goal or benchmark on what functions will be implemented-###

3.6 Testing Procedures

- Unit Testing
- System Testing
- User Acceptance Testing (UAT)

3.7 Evaluation Criteria

The system will be evaluated based on the following criteria:

- **Functionality**

Tests whether the system works as intended, such as, bookings, tracking availability, and storing information

- **Usability**

Sees how easy and user-friendly the system is for the user, and especially the owner

- **Reliability**

Assess the system's consistency to do well without errors

- **Efficiency**

How well the system is utilized

- **Security**

Ensures data safety

References:

Gupta et al. (2024), A Double-Blinded IEEE Research Paper on Car Rental System

https://ijirt.org/publishedpaper/IJIRT173737_PAPER.pdf

Divya Vanil et al. (2025) Integrated Car Booking & Rental Platform

https://www.ijesat.com/ijesat/files/V25I5029_1747217842.pdf

#####Dhoke et al. (2024) – Web Rentals: Car Rental System with Fleet Monitoring#####

#####Jyothi and Sheethal (2025). Car Rental Management System with Maintenance Automation#####

Albino, M., & Acebedo, V. (2021). Development of Car Rental Management System with Scheduling Algorithm. International Journal of Science Engineering and Technology.

https://www.ijset.in/wp-content/uploads/IJSET_V9_issue2_162.pdf

De Jesus, M. F. V., Dela Cruz, R. B. Y., Marcelo, B. F. D., Valencia, D. L., Yumul, P. K. F., & Valet, J. P. (2021). *PARAM: A Web-based Booking and Management System with Mobile Tracking Application for Param Car Rentals*. International Research Conference on Information Technology Education (IRCITE).

<https://mcc.edu.ph/academic-research/download/MCC-IT%20Research%20Journal%202020-2021.pdf>